FORM TP 2012006

JANUARY 2012

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

CHEMISTRY

Paper 032 – General Proficiency

Alternative to SBA

2 hours 10 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. Answer ALL questions on this paper.
2. Use this answer booklet when responding to the questions. For EACH question, write your answer in the space indicated and return the answer booklet at the end of the examination.
3. You may use a silent, non-programmable calculator.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Answer ALL questions.

1. (a) A student was provided with a standard solution of aqueous sodium hydroxide containing X grams of NaOH in 250 cm^3 of solution. In order to determine the mass of sodium hydroxide used, he titrated 25 cm^3 portions of the solution using 0.025 mol dm^{-3} sulphuric acid in the burette and a suitable indicator.

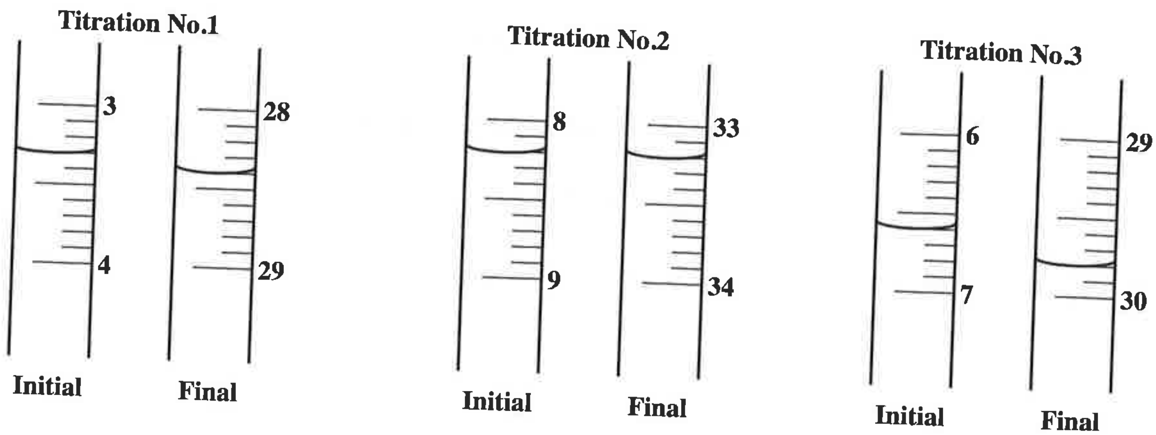

Figure 1. Burette readings showing volumes of acid used in cm^3

(i) Describe the process involved in preparing a standard solution of aqueous sodium hydroxide containing X grams in a 250 cm^3 solution.

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(3 marks)

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(ii) a) Figure 1 shows the initial and final volumes of the burette readings for the acid used. Use the information given in Figure 1 on page 2 to complete Table 1 below.

TABLE 1: RESULTS OF EXPERIMENT

Burette Readings (cm^3)	**Titration Number**		
	1	2	3
Final volume			
Initial volume			
Volume used			

(3 marks)

b) Using the best TWO values from Table 1, state the average volume of acid used in cm^3.

____

(1 mark)

(iii) a) Identify a suitable indicator for titrating sodium hydroxide and sulphuric acid.

____

____

(1 mark)

b) Explain how you will be able to determine the end-point of the reaction.

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(1 mark)

(iv) Write a **balanced** chemical equation for the reaction that occurs when sodium hydroxide reacts with dilute sulphuric acid.

____

(2 marks)

(v) Using the data that you provided in Table 1, calculate EACH of the following:

a) The number of moles of sulphuric acid used in the titration

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____

____

(1 mark)

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b) The number of moles of sodium hydroxide in 25.0 cm^3 of the solution

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(1 mark)

c) The number of moles of sodium hydroxide in 250 cm^3 of solution

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____

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(1 mark)

d) The mass of sodium hydroxide dissolved in 250 cm^3 of water
[Relative atomic masses: Na = 23; O = 16; H = 1]

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(2 marks)

(b) Another group of students conducted an experiment to copper plate an iron spoon.

(i) Identify suitable materials needed for the anode, cathode and electrolyte.

Anode: ____________________________

Cathode: ____________________________

Electrolyte: ____________________________

(2 marks)

(ii) Draw a diagram to show the arrangement of the apparatus needed to conduct this experiment.

(3 marks)

(iii) State the name of the type of reaction which occurs at the anode.

____

(1 mark)

(iv) Write a **balanced** equation to show the reaction taking place at the anode and the cathode.

Reaction at the anode:

____

Reaction at the cathode:

____

(4 marks)

Total 26 marks

2. A student conducted a number of tests on an aqueous solution of Compound Z. The inferences made are recorded in Table 2. Complete this table by filling in the observations based on the inferences recorded.

TABLE 2: TESTS ON COMPOUND Z

Test	Observations	Inferences
(a) To a sample of Solution Z, dilute nitric acid was added followed by a few drops of silver nitrate solution.	• **(1 mark)**	No Cl^-, Br- or I^- ions are formed.
(b) To a sample of Solution Z, a few copper turnings were added, followed by concentrated sulphuric acid.	• • **(2 marks)**	Nitrate ions are present.
(c) To a sample of Solution Z, aqueous sodium hydroxide was added, until in excess.	• • **(2 marks)**	Zn^{2+}, Pb^{2+} or I^- ions are present.
(d) To a sample of Solution Z, aqueous ammonia was added, until in excess.	• • **(2 marks)**	Zn^{2+} ions are confirmed.
(e) To a sample of Solution Z, a few drops of acidified aqueous potassium manganate (VII) solution were added, and the solution heated.	• **(1 mark)**	Z is not a reducing agent.
(f) To a sample of Solution Z, a few drops of barium chloride solution was added followed by dilute hydrochloric acid.	• • **(2 marks)**	SO_4^{2-} ions are present.

Total 10 marks

3. You are provided with samples of lead nitrate crystals and sodium chloride crystals. Plan and design an experiment to make lead chloride crystals. Write your responses in the spaces provided below.

(a) Procedure:

(4 marks)

(b) Apparatus and materials:

(2 marks)

(c) Precautions to be taken (this should include ONE experimental precaution and ONE safety precaution):

(2 marks)

(d) Discussion: In preparing lead chloride crystals, it is unlikely that the yield will be 100 per cent. Discuss TWO reasons why this may be so.

(4 marks)

Total 12 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.